

Edit-level Majority Voting Mitigates Over-Correction in LLM-based Grammatical Error Correction

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Over-correction issue in LLM

- LLMs tend to overcorrect.
- Why?
 - Generating unnecessary text / rewriting
 - Fine-tuning solves this issue, but is costly.

Source	I enjoyed travelling.
Unnecessary text	I enjoyed travelling. Your writing is perfect.
Unnecessary rewriting	I enjoyed traveling .

Proposal: Inference with Edit-level Majority Voting

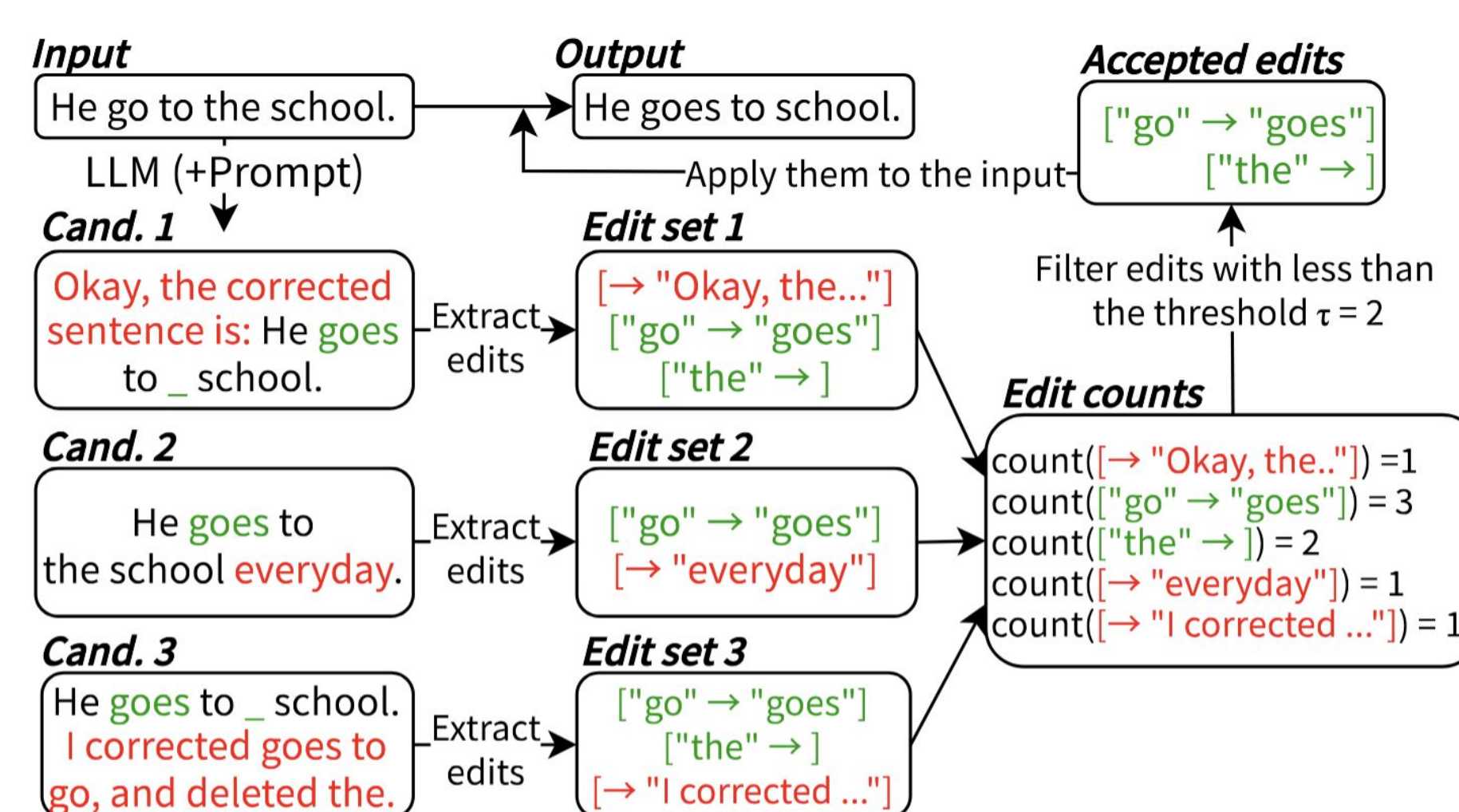
- Generate k candidates and accept edits receiving more votes than a predefined threshold (i.e., majority voting)
 - An edit is considered more reliable when it appears in more candidates
- Although majority voting is a traditional ensemble method, we apply it to a single model by generating multiple candidates

- We develop easy-to-use libraries
 - **gec-datasets**: Loading datasets
 - **gec-llm**: LLM inference

gec-datasets



gec-llm



```
1 from gec_datasets import GECDataset
2 from gec_llm.llms GECLLMZeroShot
3 from gec_llm.ensembles import EnsembleVoting
4 loader = GECDataset(base_path='sample-dir/')
5
6 model = LLMGECZeroShot(
7     model='Qwen/Qwen-8B',
8     num_candidates=8,
9 )
10 ensemble = EnsembleVoting(min_vote=7)
11 for data_id in ['bea19-dev',
12                'falko-merlin-dev', 'hi-gec-dev']:
13     # Automatically download
14     data = loader.load(data_id)
15     hyps = model.correct(data.srcs)
16     final_hyps = ensemble.ensemble(data.srcs, hyps)
```

Experiments: Effective on Minimal Edit Domain

- Evaluation on three datasets
 - Error density: CWEB-G < BEA-2019 < JFLEG
- Majority voting is effective for minimal edit.
 - It improved F0.5 on CWEB-G and BEA-2019
 - It had no effect on JFLEG-dev
→ Use cases are minimal edit. (intended!)
- No improvement was observed for a fine-tuned GEC model (EPO), as it had already mitigated overcorrection through fine-tuning.
- The proposed method achieves competitive performance against several fine-tuned models, such as GECToR (BERT), without fine-tuning.

		Minimal edit			Fluency edit						
		CWEB-G test			BEA-2019 test			JFLEG test			
Models	Inference	Prec.	Rec.	F0.5	Prec.	Rec.	F0.5	Prec.	Rec.	F0.5	GLEU
4-shot (w/o fine-tuning)											
gemma-2-9b-it	Greedy	29.4	63.9	33.0	58.4	64.4	59.5	72.5	65.0	70.8	62.6
	Proposal	40.8	52.1	42.7	65.0	60.3	64.0	69.1	66.1	68.5	62.8
Llama-3.1-8B-Instruct	Greedy	19.8	61.5	22.9	52.3	61.6	54.0	65.6	59.8	64.3	36.2
	Proposal	37.9	33.1	36.9	66.9	51.5	63.1	67.2	59.2	65.4	58.5
Fine-tuned models for GEC											
EPO (llama2-7b-chat)	Greedy	42.8	47.0	43.6	75.8	64.9	73.3	74.3	60.4	71.1	58.9
	Proposal	44.7	38.6	43.4	78.8	61.6	74.6	61.0	64.1	61.6	59.5
GECToR (BERT)	-	45.6	28.9	40.8	77.3	50.9	70.0	65.9	52.0	62.5	55.3
GECToR (RoBERTa)	-	56.1	28.3	46.9	79.3	58.0	73.9	70.8	58.6	68.0	58.9
T5-large	-	45.0	47.4	45.4	76.9	62.3	73.4	73.9	60.0	70.7	59.6

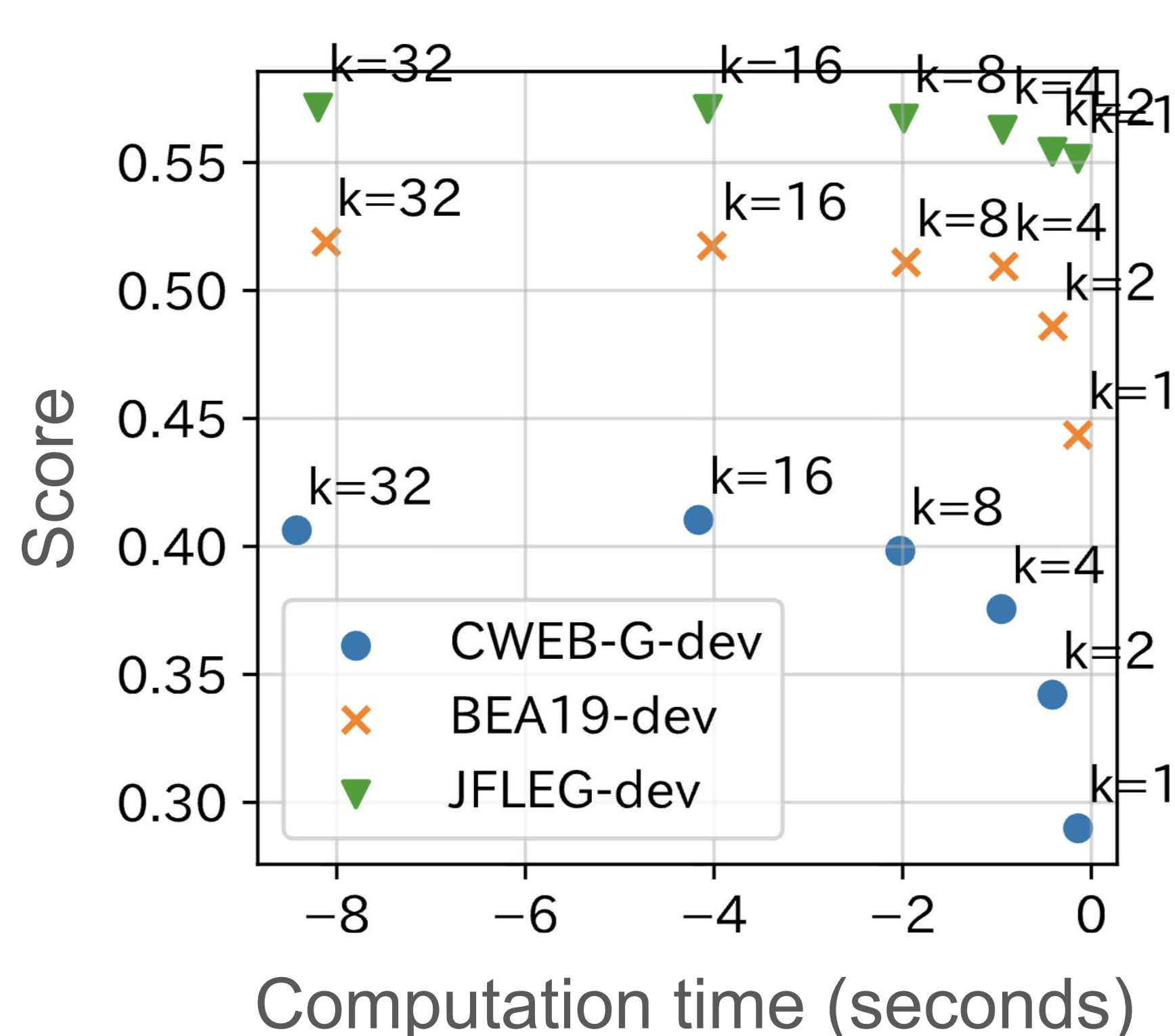
- **Bold** is the highest value for each model
- We use 8 candidates for majority voting
 - w/ top-p sampling, top_p=1.0, temperature=1.0
- We set the best threshold on development

- The paper reports results for seven languages, including English, Czech, German, Ukrainian, Korean, Hindi, and Romanian.

Discussion

Cost vs performance

- There is a trade-off
- We can adjust k



Case study

- Threshold=2,4,7 when 8 candidates
- Threshold=7 accepts only correct edits

Original	For example , when the semester start, students can not get away from the sunshine , beach , and travelling .
Reference	For example , when the semester starts , students can not get over the sunshine , beach , and travelling .
$\tau = 2$	For example , when the semester starts , students ca n't get away from the sunshine , the beaches , and traveling .
$\tau = 4$	For example , when the semester starts , students ca n't get away from the sunshine , the beach , and traveling .
$\tau = 7$	For example , when the semester starts , students can not get away from the sunshine , beach , and travelling .

Robustness in templates

- Checked avg. and stdev. in 10 different templates
- Majority voting allows for more stable output of corrected text

	CWEB-G	BEA-2019	JFLEG
Greedy	27.66 ± 3.26	43.91 ± 2.62	57.48 ± 1.01
Proposal	36.16 ± 2.85	50.68 ± 1.24	57.54 ± 0.46

Other URLs



Paper



gec-metrics
(Evaluation library)